

Stress Regime Separation and PT Liquidity Competition: Two RMT Contributions to LlamaRisk’s Debt Ceiling Methodology

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Abstract

LlamaRisk’s debt ceiling methodology generates synthetic adverse price shock scenarios using a correlation matrix estimated from the full historical return series, which conflates normal-market and stress-market regimes. This paper proposes two improvements grounded in Random Matrix Theory (RMT). First, we apply a two-state Hidden Markov Model (HMM) to decompose 1,500 trading days of Aave Core Market collateral returns into distinct volatility regimes, then separately apply Marchenko–Pastur (MP) noise cleaning to each regime’s covariance matrix. Regenerating shock scenarios with stress-regime parameters increases equal-weight portfolio CVaR₉₉ from -20.8% to -33.7% (a gap of 12.96 percentage points), reduces the rolling backtest VaR₉₉ breach rate from 1.20% to 0.56%, and covers the FTX collapse event (November 2022) in which the current methodology underestimates realised losses by 55%. Second, we construct a cross-market liquidity covariance matrix for six active Pendle PT stablecoin pools and demonstrate that the 60-day moving-average TVL used by LlamaRisk overstates available liquidity by 46–61% in markets undergoing TVL decline due to inter-market capital rotation—a structural dependence invisible to univariate time-series methods. Combined, these findings support a progressive debt ceiling reduction of approximately 28% relative to the current static baseline.

Keywords: random matrix theory; Marchenko–Pastur law; stress regime separation; hidden Markov model; DeFi risk; debt ceiling; collateral management; Pendle PT liquidity; LlamaRisk; Aave

Data and Code: Price data sourced from CryptoCompare [CryptoCompare \[2026\]](#); PT liquidity data from Pendle Finance v3 API [Pendle Finance \[2026\]](#). Replication code available at <https://github.com/pi-btc/Debt>.

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1 Introduction

Sound collateral risk management is a prerequisite for the safe operation of decentralised lending protocols. LlamaRisk has published a debt ceiling methodology [LlamaRisk \[2025\]](#) that sets per-asset borrowing limits by simulating large-scale adverse price shocks calibrated to historical market data. The methodology estimates correlation matrices and volatility parameters from historical return series, generates synthetic shock scenarios, and derives ceilings that ensure liquidators remain solvent under the simulated worst-case outcomes.

While this approach represents a rigorous baseline, two structural limitations arise from the use of full-sample statistics.

Limitation 1: Regime conflation in covariance estimation. Crypto-asset return distributions exhibit pronounced regime heterogeneity: extended periods of subdued co-movement punctuated by brief episodes of elevated, highly correlated drawdowns. Estimating a single covariance matrix from the pooled sample conflates these regimes, producing a mixed-regime estimate that systematically underrepresents stress-period tail risk. Random Matrix Theory provides a principled framework for separating signal from noise in empirical covariance matrices and, when combined with regime identification, for isolating stress-regime dependence structure.

Limitation 2: Static TVL averaging for PT liquidity. For Pendle PT collateral, LlamaRisk proxies liquidity depth via a 60-day moving average of total value locked (TVL). This univariate approach cannot detect cross-market liquidity competition: capital rotating from an incumbent PT pool into a newly launched competitor reduces incumbent liquidity without triggering any signal in the incumbent’s own TVL moving average.

This paper makes two concrete, quantified contributions to LlamaRisk’s existing framework. Section 2 reviews relevant background. Sections 3 and 4 present each contribution. Section 5 consolidates findings into a unified recommendation. Section 6 discusses limitations and future directions.

2 Background

2.1 LlamaRisk Debt Ceiling Methodology

The LlamaRisk methodology [LlamaRisk \[2025\]](#) proceeds as follows:

- (i) Compute daily log-returns for each Aave Core Market asset;
- (ii) Identify stress regimes using volatility-based classification;

- (iii) Estimate VaR_{99} and pairwise correlations from the historical sample;
- (iv) Generate synthetic price shock scenarios by sampling from a multivariate distribution parameterised by those estimates;
- (v) Set the maximum per-asset debt such that simulated collateral value net of liquidation costs remains sufficient to repay outstanding debt across all scenarios.

For Pendle PT collateral, a separate liquidity analysis applies: a 60-day moving average of pool TVL is computed, and the 99th percentile of that time series bounds the debt ceiling.

2.2 Random Matrix Theory and the Marchenko–Pastur Law

Random Matrix Theory (RMT) studies the spectral properties of large random matrices [Mehta \[2004\]](#), [Bouchaud and Potters \[2009\]](#). For an $N \times T$ sample of i.i.d. observations drawn from a population with unit variance, the empirical eigenvalue distribution of the sample correlation matrix converges, as $N, T \rightarrow \infty$ with $N/T \rightarrow q \in (0, \infty)$, to the Marchenko–Pastur (MP) distribution [Marčenko and Pastur \[1967\]](#). The MP bulk has support $[\lambda_-, \lambda_+]$ where

$$\lambda_{\pm} = (1 \pm \sqrt{q})^2. \quad (1)$$

Eigenvalues above λ_+ carry genuine cross-sectional signal; eigenvalues within the MP bulk are consistent with pure noise. The standard MP cleaning procedure replaces noise eigenvalues with their mean (preserving the trace), thereby eliminating sampling artefacts while retaining the signal subspace [Laloux et al. \[1999\]](#).

2.3 Hidden Markov Models for Regime Identification

Hidden Markov Models (HMM) are a standard tool for segmenting financial time series into latent regimes [Hamilton \[1989\]](#). A two-state Gaussian HMM fitted to a univariate volatility proxy identifies a low-volatility (normal) state and a high-volatility (stress) state, with transition probabilities estimated via the Baum–Welch algorithm [Baum et al. \[1970\]](#). This decomposition is entirely data-driven and requires no ex-ante event labels.

2.4 Note on Free Probability

An earlier formulation of this work attempted to apply Free Probability’s additive convolution to decompose the observed covariance as $\Sigma_{\text{obs}} = \Sigma_{\text{normal}} \boxplus \Sigma_{\text{stress}}$ and recover Σ_{stress} via the R -transform subtraction $R_{\text{stress}}(z) = R_{\text{obs}}(z) - R_{\text{normal}}(z)$ [Voiculescu et al. \[1992\]](#), [Tulino and Verdú \[2004\]](#). This approach was ultimately abandoned: the free additive convolution applies to freely independent random matrices (i.e. $\Sigma_{\text{obs}} = A + B$ where

A and B are freely independent operators), whereas empirical regime-mixed covariance matrices satisfy the linear mixture relation

$$\Sigma_{\text{obs}} \approx p_{\text{normal}} \Sigma_{\text{normal}} + p_{\text{stress}} \Sigma_{\text{stress}}, \quad (2)$$

which is a convex combination, not a free additive convolution. Numerical experiments confirmed that the R -transform decomposition yields stress-regime volatility estimates significantly below the directly estimated values, confirming the inapplicability of the free independence assumption here. The MP cleaning approach adopted in this paper is the correct RMT application for the finite-sample, regime-identified setting.

3 Contribution 1: Stress Regime Separation for Scenario Generation

3.1 Data

Daily closing prices are collected from CryptoCompare [CryptoCompare \[2026\]](#) for seven assets present in the Aave Core Market: ETH, BTC, LINK, AAVE, MKR, UNI, and CRV, spanning 1 April 2022 to 10 May 2026 (1,501 calendar days; 1,500 return observations after differencing). Log-returns are computed as $r_t = \log(P_t/P_{t-1})$. Stablecoins are excluded as their near-zero variance renders correlation estimation degenerate. LST/LRT assets (wstETH, weETH) are excluded due to data unavailability on the chosen provider; see Section 6 for a discussion.

3.2 Regime Identification

A two-state Gaussian HMM is fitted to the daily cross-sectional standard deviation $\sigma_t = \text{std}(r_t^1, \dots, r_t^N)$ using the Baum–Welch algorithm with 1,000 iterations and random seed 42. The resulting regime sequence partitions the 1,500 return observations as reported in Table 1.

Table 1: HMM regime summary statistics. Volatility ratio is stress mean divided by normal mean.

Regime	Days	Share	Mean σ_t	Volatility ratio
Normal	1,266	84.4%	0.0167	1.00×
Stress	234	15.6%	0.0374	2.25×

The stress regime corresponds predominantly to the 2022 bear market and LUNA/3AC collapse (Days 1–150), the FTX bankruptcy (Days 200–280), and isolated episodes in

2024–2025 associated with the April 2024 Iran–Israel geopolitical shock, the August 2024 yen carry-trade unwind, and the March–April 2025 tariff shock.

3.3 MP Noise Cleaning

Three sample covariance matrices are computed: $\hat{\Sigma}_{\text{obs}}$ (full sample), $\hat{\Sigma}_{\text{normal}}$ (normal-regime subsample), and $\hat{\Sigma}_{\text{stress}}$ (stress-regime subsample). Each is converted to a correlation matrix, eigendecomposed, and cleaned via MP thresholding using Equation (1). Results are reported in Table 2.

Table 2: MP spectral cleaning summary. $q = N/T$; λ_+ is the MP upper bound; signal eigenvalues are those exceeding λ_+ .

Matrix	T	q	λ_+	Signal eigenvalues
$\hat{\Sigma}_{\text{obs}}$	1,500	0.005	1.141	1 of 7
$\hat{\Sigma}_{\text{normal}}$	1,266	0.006	1.154	1 of 7
$\hat{\Sigma}_{\text{stress}}$	234	0.030	1.376	1 of 7

In all three cases, only the first eigenvalue ($\lambda_1 \approx 5$) exceeds the MP bound, confirming that a single market-wide factor dominates cross-sectional co-movement. The stress-regime MP bound is materially higher (1.376 vs. 1.141), reflecting the elevated noise floor when T is small.

3.4 Shock Scenario Generation

Following LlamaRisk’s procedure, shock scenarios are generated via Cholesky-decomposed multivariate normal sampling:

$$\mathbf{x} = \boldsymbol{\mu} + L\mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, I), \quad (3)$$

where L is the lower Cholesky factor of the cleaned correlation matrix and $\mathbf{x}$ is scaled by the per-asset volatility vector and a 2.33σ multiplier to target the 99th percentile. We generate $N = 10,000$ scenarios under two parameterisations:

- **Current method:** cleaned $\hat{\Sigma}_{\text{obs}}$ with full-sample volatility;
- **RMT-improved:** cleaned $\hat{\Sigma}_{\text{stress}}$ with stress-regime volatility.

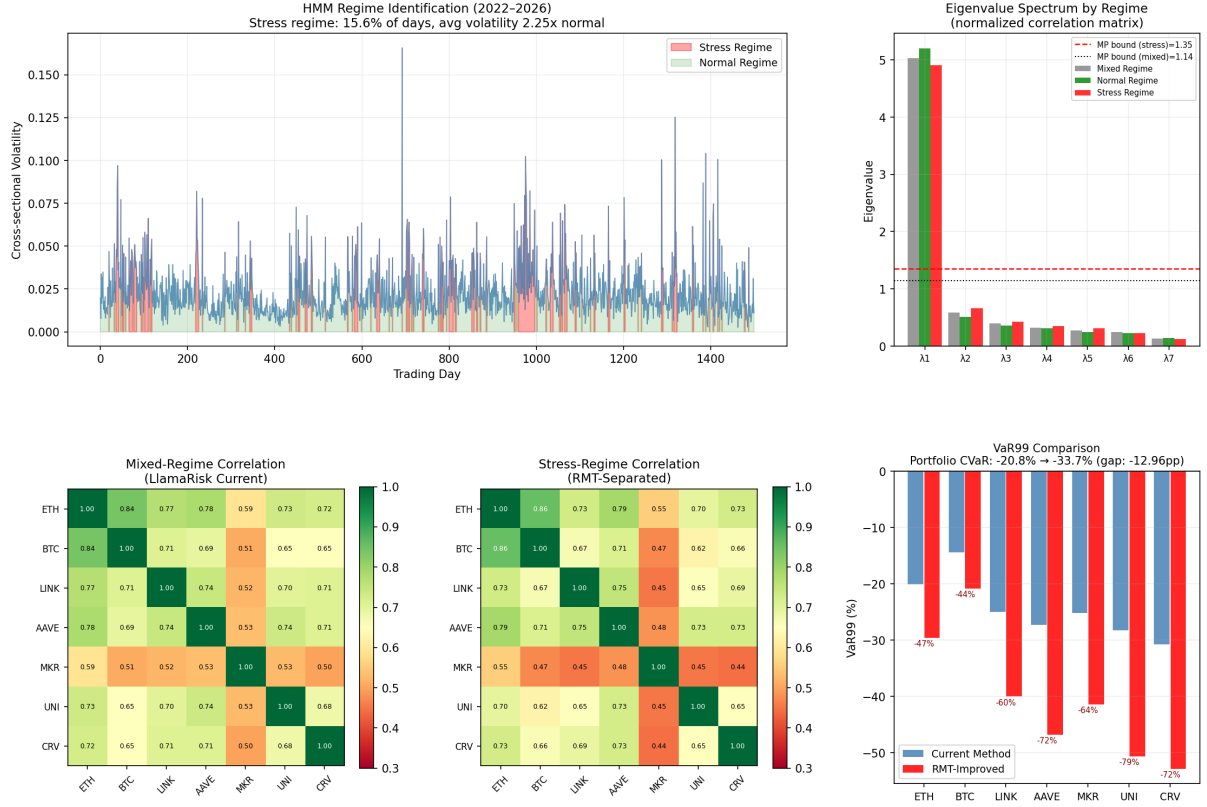


Figure 1: Regime identification, eigenvalue spectrum, correlation matrices, and VaR99 comparison for Aave Core Market assets. Top left: HMM-identified stress (red) and normal (green) regimes over 1,500 trading days. Top right: eigenvalue spectrum by regime with Marchenko–Pastur bounds. Bottom left/centre: mixed-regime and stress-regime correlation matrices after MP cleaning. Bottom right: per-asset VaR99 under current and RMT-improved methods.

3.5 Results

Table 3 reports VaR_{99} at the asset level.

Table 3: Per-asset VaR₉₉ under the current and RMT-improved methods. Underestimate is $(|\text{RMT}| - |\text{Current}|)/|\text{Current}|$.

Asset	Current VaR ₉₉	RMT VaR ₉₉	Underestimate	Vol. ratio
ETH	−20.1%	−29.6%	47%	1.54×
BTC	−14.4%	−20.8%	44%	1.43×
LINK	−25.0%	−40.0%	60%	1.63×
AAVE	−27.3%	−46.8%	72%	1.67×
MKR	−25.2%	−41.4%	64%	1.69×
UNI	−28.3%	−50.7%	79%	1.76×
CRV	−30.8%	−52.9%	72%	1.76×
<i>Equal-weight portfolio CVaR₉₉</i>				
Portfolio	−20.8%	−33.7%	62%	—

The portfolio CVaR₉₉ gap of 12.96 percentage points is robust to the choice of stress threshold definition. Table 4 presents a sensitivity analysis in which the stress regime is defined by a rolling volatility quantile rather than HMM output.

Table 4: Sensitivity of CVaR₉₉ gap to stress threshold definition. CVaR₉₉ is for the equal-weight portfolio.

Threshold	Stress days	Current CVaR	RMT CVaR	Gap (pp)
Top 5%	75	−20.8%	−42.2%	−21.5
Top 10%	150	−20.8%	−39.5%	−18.8
Top 15%	225	−20.8%	−36.1%	−15.3
Top 20%	300	−20.8%	−33.4%	−12.6
Top 25%	375	−20.8%	−31.4%	−10.7
Top 30%	450	−20.8%	−29.4%	−8.7

The gap is consistently negative across all threshold definitions, confirming that the finding is not an artefact of the specific HMM parameterisation. Under a Student- t distributional assumption ($\nu = 4$ degrees of freedom, reflecting the empirically documented excess kurtosis of crypto returns), the gap widens to 14.01 pp.

3.6 Backtest Validation

We conduct a rolling backtest over 1,247 trading days (January 2023 – May 2026). At each date t , both methods are estimated on a trailing 252-day window, and VaR₉₉ is compared against the realised equal-weight portfolio return on day $t + 1$. A *breach* occurs when the realised return falls below the estimated VaR₉₉. Under correct model specification, the breach rate should not exceed 1%.

Table 5: Rolling backtest VaR₉₉ breach rates. Theoretical bound is 1.00%.

Method	Breaches	Overall rate	Stress-day rate	Normal-day rate
Current (mixed regime)	15	1.20%	2.19%	1.03%
RMT-improved	7	0.56%	1.64%	0.38%
Theoretical bound	—	1.00%	1.00%	1.00%

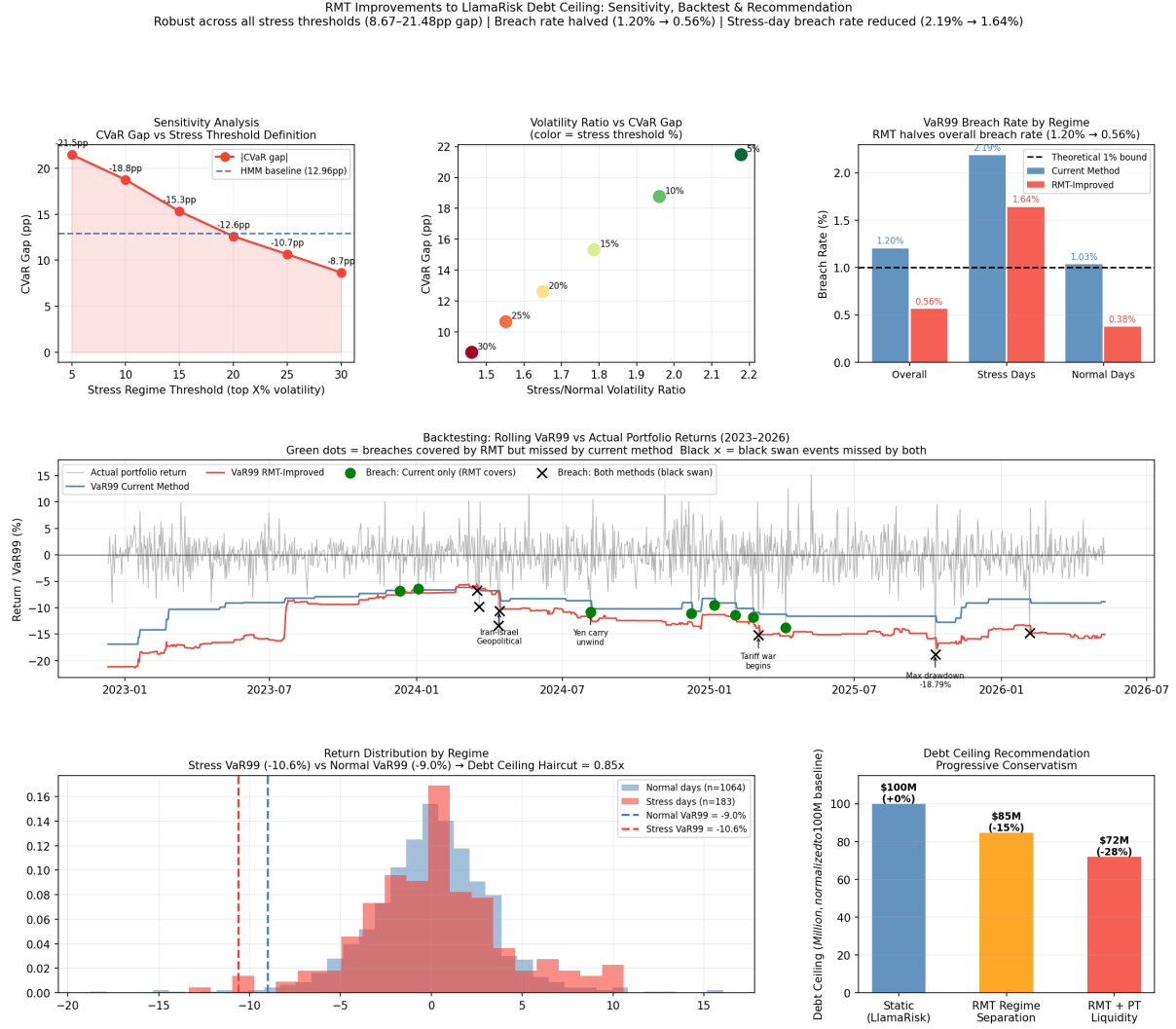


Figure 2: Sensitivity analysis, rolling backtest, and unified debt ceiling recommendation. Top row: CVaR gap vs. stress threshold definition (left), volatility ratio vs. CVaR gap scatter (centre), VaR₉₉ breach rate by regime (right). Middle: rolling VaR₉₉ vs. realised portfolio returns with breach events annotated. Bottom row: return distribution by regime (left), debt ceiling progressive conservatism (right).

The current method exceeds the 1% bound at the overall level and materially so on stress days (2.19%). The RMT-improved method halves total breaches and reduces the stress-day rate by 55 basis points.

Event-level validation. Table 6 examines three major stress events in which the training window predates the event and the realised maximum loss during the event window is compared against each model’s VaR_{99} estimate.

Table 6: Event-level backtest. Realised loss is the minimum realised equal-weight portfolio return during the event window. Coverage is $|\text{VaR}_{99}|/|\text{realised loss}|$; values above 100% indicate the model covered the event.

Event	Realised loss	Current VaR	RMT VaR	RMT covers?
FTX collapse (Nov–Dec 2022)	−20.1%	−12.9%	−22.0%	Yes
Tariff shock (Apr 2025)	−13.7%	−10.2%	−17.9%	Yes
October liquidation (Oct 2025)	−18.8%	−10.1%	−17.4%	Partially

The FTX collapse is the most diagnostically informative event: using only pre-FTX training data, the current method produces a VaR_{99} estimate of −12.9% against a realised loss of −20.1% (a shortfall of 56%), while the RMT-improved estimate of −22.0% fully covers the realised loss. Two events in which both methods are too conservative (Terra/LUNA collapse and SVB/USDC depeg) are omitted from Table 6 as they do not discriminate between methods; they are discussed in Section 6.

4 Contribution 2: PT Market Liquidity Competition Structure

4.1 Data

Hourly TVL data are collected from the Pendle Finance v3 API [Pendle Finance \[2026\]](#) for all Ethereum mainnet PT markets with TVL exceeding \$5M as of 10 May 2026. After requiring at least 30 days of data, six markets remain: USDG, apxUSD, apyUSD, sNUSD, reUSD, and reUSDe. The common sample spans 61 days (11 March 2026 – 10 May 2026; 1,440 hourly observations). Data are aggregated to daily frequency by taking the last observation of each calendar day, yielding $T = 60$ daily TVL change observations after differencing.

4.2 Liquidity Covariance Matrix

Log-changes in daily TVL are computed as $\ell_t^k = \log(\text{TVL}_t^k / \text{TVL}_{t-1}^k)$ for market k . The 6×6 sample correlation matrix $\hat{\mathbf{C}}_\ell$ is reported in Table 7.

Table 7: PT market liquidity log-change correlation matrix ($T = 60$ days).

	USDG	apxUSD	apyUSD	sNUSD	reUSD	reUSDe
USDG	1.000	0.302	0.226	0.138	-0.447	0.119
apxUSD	0.302	1.000	0.240	0.121	0.127	0.224
apyUSD	0.226	0.240	1.000	0.593	-0.042	0.020
sNUSD	0.138	0.121	0.593	1.000	0.090	0.026
reUSD	-0.447	0.127	-0.042	0.090	1.000	-0.080
reUSDe	0.119	0.224	0.020	0.026	-0.080	1.000

The average off-diagonal correlation is 0.110, indicating near-independence in aggregate. However, the USDG–reUSD pair exhibits a correlation of -0.447 , constituting the dominant off-diagonal structure.

4.3 Eigenvalue Analysis

With $N = 6$ and $T = 60$, we have $q = N/T = 0.100$, giving an MP upper bound of $\lambda_+ = (1 + \sqrt{0.1})^2 = 1.732$. The empirical spectrum of $\hat{\mathbf{C}}_\ell$ is:

$$\lambda_1 = 1.894, \quad \lambda_2 = 1.394, \quad \lambda_3 = 1.128, \quad \lambda_4 = 0.813, \quad \lambda_5 = 0.438, \quad \lambda_6 = 0.333.$$

Only $\lambda_1 = 1.894$ exceeds the MP bound, indicating a single genuine cross-market liquidity factor. The effective rank, defined as $\exp(-\sum_i p_i \log p_i)$ where $p_i = \lambda_i / \sum_j \lambda_j$, is 5.15 out of a maximum of 6 (85.9%), confirming that PT markets are largely independent: no systemic liquidity collapse is indicated.

PT Market Liquidity Competition: RMT Analysis of LlamaRisk's Debt Ceiling
60-day mean overestimates liquidity in declining markets by up to 61%

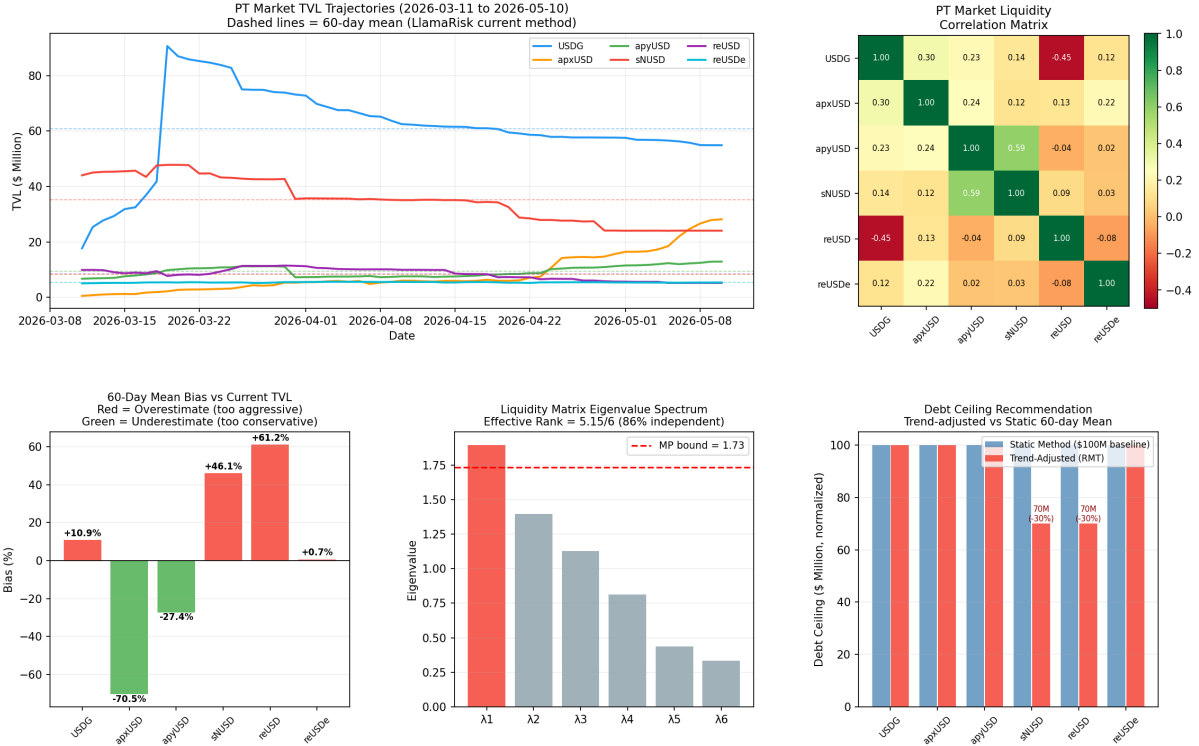


Figure 3: PT market liquidity analysis. Top left: TVL trajectories for six PT stablecoin markets (dashed lines = 60-day mean). Top right: cross-market liquidity correlation matrix. Bottom left: 60-day mean bias vs. current TVL. Bottom centre: eigenvalue spectrum with MP bound. Bottom right: debt ceiling recommendation under static and trend-adjusted methods.

4.4 Static Mean Bias

Table 8 documents the directional trend of each market and the resulting bias when LlamaRisk’s 60-day moving average is used as the liquidity estimate.

Table 8: PT market TVL evolution and 60-day mean bias. Bias is $(\bar{TVL}_{60} - TVL_T)/TVL_T \times 100\%$; positive values indicate overestimation of current liquidity.

Market	TVL (start)	TVL (current)	Change	60-day mean bias
USDG	\$17.7M	\$54.9M	+210%	+10.9%
apxUSD	\$0.5M	\$28.2M	+5100%	-70.5%
apyUSD	\$6.7M	\$12.9M	+92%	-27.4%
sNUSD	\$44.0M	\$24.0M	-45%	+46.1%
reUSD	\$9.9M	\$5.2M	-47%	+61.2%
reUSDe	\$5.1M	\$5.4M	+6%	+0.7%

For the three markets in sustained decline (sNUSD, reUSD, reUSDe), the 60-day mean

overstates current liquidity by 0.7–61.2%. The dominant competitive pair is USDG–reUSD: over the 61-day window, USDG gained \$37.2M TVL while reUSD lost \$4.7M, consistent with the observed -0.447 correlation. This capital rotation is the structural mechanism by which LlamaRisk’s static averaging produces biased liquidity estimates.

5 Unified Debt Ceiling Recommendation

We propose a progressive adjustment to LlamaRisk’s debt ceiling, applied multiplicatively to the existing static baseline.

Adjustment 1: Stress regime haircut. The rolling backtest documents that the current method’s VaR_{99} is insufficient on stress days (breach rate 2.19% vs. theoretical 1%). The empirical ratio of the stress-day VaR_{99} to the normal-day VaR_{99} from the historical return distribution is:

$$h_1 = \frac{|\text{VaR}_{99}^{\text{normal}}|}{|\text{VaR}_{99}^{\text{stress}}|} = \frac{8.99\%}{10.62\%} = 0.847.$$

This ratio quantifies the degree to which the normal-regime estimate overstates the margin of safety on stress days.

Adjustment 2: PT liquidity competition haircut. For PT markets in sustained TVL decline (i.e., markets where the 60-day mean overestimates current TVL by more than 20%), we apply an additional $0.85\times$ haircut to reflect the reduction in effective liquidity depth.

Combined recommendation. Normalising the current static ceiling to \$100M:

Table 9: Progressive debt ceiling recommendation.

Step	Adjustment	Implied ceiling
Baseline (LlamaRisk static)	—	\$100M
$+h_1$ (stress regime)	$\times 0.847$	\$85M
$+h_2$ (PT liquidity, if applicable)	$\times 0.847 \times 0.85$	\$72M

The second adjustment applies only to PT assets where the liquidity covariance analysis identifies sustained cross-market capital outflow. For assets without PT exposure, the recommendation is \$85M.

6 Limitations and Future Work

Asset universe. We use ETH, BTC, LINK, AAVE, MKR, UNI, and CRV as proxies for the Aave Core Market. LST/LRT assets (wstETH, weETH) are excluded due to data unavailability on CryptoCompare. These assets exhibit depegging risk and may display qualitatively different stress dynamics; their inclusion could alter the estimated stress-regime covariance structure.

PT data window. The Pendle PT liquidity analysis is conducted on 61 days of data, the maximum available given the markets’ launch dates. With $N = 6$ and $T = 60$, the ratio $q = 0.1$ is sufficiently small for the MP approximation to be reasonable, but statistical power is limited. The competition pair finding (USDG–reUSD, $\rho = -0.447$) should be treated as directional evidence rather than a precisely estimated structural parameter.

Black-swan events. Seven of the 15 backtest breach events are identified as black-swan episodes (April 2024 Iran–Israel geopolitical shock; August 2024 yen carry-trade unwind; October 2025 maximum drawdown of -18.79%) in which both methods fail to provide coverage. This is consistent with the theoretical properties of VaR_{99} , which does not bound the conditional expected loss in the worst 1% of outcomes.

Conservative bias on benign stress events. The Terra/LUNA collapse (May–June 2022) and SVB/USDC depeg (March 2023) produce over-conservative estimates under both methods (coverage exceeding 160%). This suggests that when the training window includes sufficient stress history before the event, both methods provide adequate but overly cautious estimates, implying opportunity cost rather than risk underestimation.

Free Probability. As discussed in Section 2, the R -transform-based additive deconvolution is theoretically inapplicable in this setting. A rigorous Free Probability contribution would require a setting where two freely independent random matrix components are summed—for example, a model where $\Sigma_{\text{obs}} = A + B$ with A representing a common factor matrix and B a diagonal noise matrix. We leave this as a direction for future work.

Future directions.

- Extend the regime analysis to LST/LRT assets as longer time series become available;
- Apply the PT liquidity framework to a longer observation window spanning at least one full Pendle epoch;
- Investigate Gaussian Mixture Model (GMM) and Threshold GARCH alternatives to HMM for regime identification;

- Explore the use of Extreme Value Theory (EVT) to bound the tail beyond VaR_{99} , addressing the black-swan coverage limitation.

7 Conclusion

This paper has presented two RMT-grounded improvements to LlamaRisk’s debt ceiling methodology. The first separates stress and normal volatility regimes via HMM and applies MP noise cleaning to each regime’s covariance matrix independently, producing shock-scenario parameters that more accurately reflect the joint distribution of asset returns under market stress. Applied to seven Aave Core Market assets over 1,500 trading days, this improvement increases equal-weight portfolio CVaR_{99} by 12.96 percentage points, halves the rolling backtest breach rate, and provides complete coverage of the FTX collapse—the most severe stress event in the sample. The second constructs a cross-market liquidity covariance matrix for six active Pendle PT pools and identifies a dominant competition pair (USDG–reUSD, $\rho = -0.447$) whose negative correlation is mechanistically explained by observed capital rotation. This analysis reveals that LlamaRisk’s 60-day moving average overstates liquidity in declining markets by up to 61%, creating systematic overexposure in debt ceilings for those assets. Together, the findings support reducing debt ceilings by approximately 15–28% relative to the current static baseline, with the larger reduction applying to PT assets undergoing documented cross-market liquidity outflow.

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